One Day or Three? Hong Kong and Kazakhstan Equities After U.S. Sell-Offs

Independent Research Project

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Chapter 1 Introduction

How do stock markets in Hong Kong and Kazakhstan perform following a sharp decline in the U.S. stock market, and does their relative performance change when the observation window extends from one to three local trading days? Comparing multiple horizons allows the analysis to distinguish an initial return gap from a subsequent change in that gap. Trading calendars also matter: a local market holiday can leave more than one U.S. trading session between the previous local close and the next local open, complicating comparisons across markets. This study examines these issues by comparing the cumulative returns of the Hang Seng Index and Kazakhstan's KASE Index following sharp declines in the S&P 500.

The primary research question is whether the average cumulative return gap between the two markets changes from the first to the third local trading day. The gap is defined as the cumulative return of the KASE Index minus that of the Hang Seng Index, expressed in percentage points. For each market, the baseline is the most recent local closing value before the U.S. event trading session begins. Cumulative returns are measured from this baseline to the close of the first or third local trading session following the U.S. event close. The study estimates the average gap at each horizon and directly evaluates its change by subtracting the Day 1 gap from the Day 3 gap for each event.

The analysis uses daily data for the three indices from 2017 through 2025. In the main specification, a sharp U.S. market decline is defined as an S&P 500 daily return of −2% or lower. A calendar filter retains events only when exactly one U.S. trading session falls between the baseline close and the Day 1 open in each local market. This rule retains 65 of the 85 candidate events. The retained events form 47 groups of overlapping observation windows, which serve as the resampling units for the cluster bootstrap. The study also compares the return gaps with a quiet-day benchmark weighted to match the distribution of retained decline events across years. Uncertainty in this benchmark comparison is assessed using a block bootstrap. All returns are calculated from local-currency price indices.

The estimated average KASE–HSI return gap on Day 1 is 0.669 percentage points, with a 95% cluster-bootstrap confidence interval of (0.345, 1.003). By Day 3, the estimated gap is 0.459 percentage points, with a corresponding confidence interval of (-0.441, 1.177). The estimated change from Day 1 to Day 3 is −0.210 percentage points, with a confidence interval of (−0.995, 0.391). Although the point estimates suggest a smaller positive gap at the longer horizon, the interval for the change includes zero. The evidence therefore supports a positive average Day 1 gap but does not establish that the gap narrows by Day 3.

Relative to the weighted quiet-day benchmark, the Day 1 gap is larger by 0.606 percentage points, with a 95% block-bootstrap confidence interval of (0.209, 1.002), using a block length of 20 U.S. trading sessions. This finding indicates that the observed Day 1 difference exceeds the corresponding benchmark difference in the selected quiet-day sample. The comparison remains conditional on the event and benchmark selection rules and does not identify a causal effect of U.S. market declines.

Chapter 2 Related Literature

Chen, Yang, and Chang (2025) examine return and volatility spillovers among stock markets in the United States, Hong Kong and mainland China. They identify the United States as a major transmitter of returns and volatility and Hong Kong as a major recipient. They also report that these relationships vary over time and across market conditions, including financial crises, the COVID-19 pandemic, and U.S.–China trade tensions. Their findings provide context for examining Hong Kong's performance following sharp U.S. market declines. The present study focuses on cumulative returns following selected decline events and compares Hong Kong's performance with Kazakhstan's over one and three local trading days.

Duskaliyeva and Belgibayev (2023), in a working paper published by the National Bank of Kazakhstan, investigate the relationship between Kazakhstan's stock market returns and macroeconomic variables. Their monthly model includes S&P 500 returns as an indicator of the external market environment and compares periods before and during inflation targeting. They report that the estimated relationship with the external market weakened during inflation targeting, while noting that several model estimates were statistically insignificant. Their analysis provides context for examining KASE returns in relation to domestic macroeconomic conditions and the external market environment. The present study adds a daily-frequency perspective by examining KASE's cumulative returns over the first three local trading days following sharp U.S. declines.

Maderitsch (2015) examines information transmission among Hong Kong, European, and U.S. stock markets using high-frequency data and non-overlapping intraday returns. The study finds temporary return spillovers and reports that some relationships depend on the preceding market's volatility state. Its chronological treatment of returns highlights the importance of identifying which trading period precedes another. With daily data, matching calendar dates alone can produce unequal information windows around holidays. The present study therefore defines observation horizons using each market's local trading days and retains events with exactly one U.S. session between each market's baseline close and Day 1 Open.

MacKinlay (1997) provides a methodological framework for event studies, covering event-window selection, normal-return models, and abnormal-return measurement. In this framework, abnormal returns measure the difference between observed returns and an estimated normal-return benchmark. The present study organizes observations around specified events. Its principal measure is the cumulative return gap, calculated as KASE's actual cumulative return minus the Hang Seng Index's actual cumulative return. A separately constructed quiet-day sample provides an additional empirical benchmark, weighted to match the annual distribution of decline events. The resulting measures represent cumulative return comparisons, while the quiet-day adjustment measures the difference between the selected sell-off and quiet-day conditions.

The present study examines the paired cumulative returns of Hong Kong and Kazakhstan following the same U.S. sell-off events during 2017–2025. It combines event selection that accounts for trading-calendar differences with a direct estimate of the change in the KASE–HSI return gap from Day 1 to Day 3. The quiet-day comparison provides an additional benchmark for interpreting this gap. Resampling procedures and sensitivity checks assess the uncertainty and robustness of these comparisons.

References

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Chapter 3 Data and Sample Construction

3.1 Data Sources and Coverage

This study uses daily closing values for the S&P 500, the Hang Seng Index (HSI), and the Kazakhstan Stock Exchange Index (KASE) over 2017–2025. S&P 500 data were obtained from the Federal Reserve Economic Data (FRED) database, and HSI data were obtained from Yahoo Finance. KASE historical data were downloaded from the Kazakhstan Stock Exchange and extracted from an archived Excel file.

The S&P 500 is used to identify sharp U.S. market declines, while the HSI and KASE Index measure subsequent market performance in Hong Kong and Kazakhstan. Returns are calculated from local-currency price indices. These measures exclude exchange-rate movements, dividends, and transaction costs.

3.2 Data Preparation and Return Calculation

Dates were converted to a consistent format, and observations in each series were sorted chronologically. For the KASE archive, the date and closing-value columns were retained for analysis. Daily returns were calculated as percentage changes between consecutive trading-day closes:

$$r_{i,t} = \left(\frac{P_{i,t}}{P_{i,t-1}} - 1 \right) \times 100,$$

where $P_{i,t}$ denotes the closing value of index i on its trading date t and $P_{i,t-1}$ denotes its preceding available trading-day close.
The first observation in each series has no calculated return because a preceding closing value is unavailable.
A U.S. sell-off event is identified whenever the S&P 500 daily return is −2% or lower. For each event, cumulative returns in Hong Kong and Kazakhstan are measured from the respective baseline close to the close of the first and third local trading sessions following the U.S. event close. The baseline is the most recent local close before the U.S. event session begins. Each market follows its own trading calendar, so Day 3 may fall on different calendar dates across the two markets.

3.3 Event Matching and Calendar Filter

The initial search identified 85 qualifying S&P 500 decline events during 2017–2025. Each event was matched to a baseline close and subsequent trading sessions in both local markets. Differences in exchange holidays can allow multiple U.S. trading sessions to occur between a local baseline close and the next local opening, complicating comparisons of the first observed local response.

The main sample therefore retains an event only when exactly one U.S. trading session occurs between the baseline close and the Day 1 open in each local market. Applying this requirement to both markets retains 65 events and excludes 20, equivalent to approximately 23.5% of the candidate sample. This filter reduces differences in the number of U.S. sessions preceding the first observed local response.

3.4 Overlapping Event Windows

Some retained events occur close enough for their three-day observation windows to overlap. Such overlap creates dependence because different event observations can share local-market trading days and price movements.

All 65 retained events remain in the analysis. Events with overlapping windows are assigned to common resampling groups, producing 47 overlap groups. The cluster bootstrap resamples these groups to account for dependence associated with overlapping windows. Further details of the grouping and resampling procedures are provided in Chapter 4.

Table 1. Event sample construction

Sample stage	Count	Unit
Candidate S&P 500 decline events	85	Events
Excluded by the calendar filter	20	Events
Retained in the main sample	65	Events
Overlapping-window groups among retained events	47	Groups

Chapter 4 Methodology

4.1 Event Definition and Calendar Alignment

A U.S. sell-off event is defined as a trading day on which the S&P 500 daily return is −2% or lower. For each event, the baseline for each local market is the last available closing value dated on or before the U.S. event date. Day 1, Day 2, and Day 3 are the first three available local trading dates strictly after that date. Both markets must have a baseline and a complete three-session observation window.

Calendar alignment is implemented using trading dates. For each market, the analysis counts the U.S. trading dates from the baseline date, inclusive, to the Day 1 date, exclusive. An event is retained only when this count equals one in both markets. Applying this rule retains 65 of the 85 candidate events.

The filter limits differences in the number of U.S. trading sessions covered before the first subsequent local trading day. Its interpretation depends on the chronological ordering of trading sessions across markets. Matching uses daily dates, and the elapsed time to each horizon can differ between Hong Kong and Kazakhstan.

4.2 Cumulative Returns and the Return Gap

For market m, event i, and local trading-day horizon h, cumulative return is calculated as

$$R_{m,i,h} = \left(\frac{P_{m,i,h}}{P_{m,i,0}} - 1 \right) \times 100,$$

where $P_{m,i,0}$ is the baseline closing value and $P_{m,i,h}$ is the closing value on local trading day h. The principal horizons are $h = 1$ and $h = 3$. Returns are expressed as percentages and measured using local-currency price indices.

The paired cumulative return gap is defined as

$$D_{i,h} = R_{\mathrm{KASE},i,h} - R_{\mathrm{HSI},i,h}.$$

The gap is expressed in percentage points. A positive value indicates that KASE recorded a higher cumulative return than the HSI over the corresponding event window. The sample mean at each horizon is

$$\overline{D}_h = \frac{1}{N} \sum_{i=1}^{N} D_{i,h},$$

where N is the number of retained events.

The primary comparison concerns the change in the signed return gap from Day 1 to Day 3. Its sample estimate is

$$\overline{\Delta D} = \frac{1}{N} \sum_{i=1}^{N} \Delta D_i = \overline{D}_3 - \overline{D}_1$$

A negative value indicates a decline in the signed KASE–HSI gap. Convergence toward zero is a separate property, measured at the event level by a reduction in $|D_{i,h}|$. Inference on the cross-horizon change uses the distribution of $\overline{\Delta D}$ directly.

4.3 Overlapping Windows and Cluster Bootstrap

For clustering purposes, each event's observation window extends from the earlier of the two markets' baseline dates to the later of their Day 3 dates. Events are assigned to the same cluster

when these intervals overlap or share an endpoint. Clusters also include events connected indirectly through overlapping intermediate windows. The 65 retained events form 47 clusters.

Each bootstrap repetition draws 47 clusters with replacement and includes all events belonging to each selected cluster. The paired HSI and KASE outcomes remain together. The mean Day 1 gap, mean Day 3 gap, and mean gap change are recalculated from the sampled events. Each sampled event receives equal weight, so the denominator is the total number of events contained in the sampled clusters.

The procedure uses 10,000 repetitions and a random seed of 42. The 2.5th and 97.5th percentiles of each bootstrap distribution define its 95% confidence interval. This procedure addresses dependence within the constructed overlap groups. Its interpretation relies on the adequacy of the grouping and the assumptions underlying cluster resampling.

4.4 Quiet-Day Benchmark

A quiet day is a U.S. trading day on which the absolute S&P 500 daily return is no greater than 0.5%. Eligible quiet days must satisfy the same calendar filter and complete-window requirements as sell-off events. Dates within three U.S. trading sessions before or after any raw −2% sell-off are excluded. Quiet-day windows that overlap or share an endpoint with any of the 85 raw sell-off windows are also excluded.

The comparison uses years represented in the retained sell-off sample and contains 672 eligible quiet-day observations. Annual quiet-day mean gaps are weighted according to the annual distribution of retained sell-off events. Let n_y denote the number of retained sell-off events in year y , and let $\overline{D}_{Q,y,h}$ denote that year's mean quiet-day gap. The weighted benchmark is

$$\overline{D}^{w}_{Q,h} = \sum_{y \in \mathcal{Y}} \frac{n_y}{N} \overline{D}_{Q,y,h}$$

where $\mathcal{Y}$ is the set of comparison years. The benchmark-adjusted gap and its cross-horizon change are

$$E_h = \overline{D}_{S,h} - \overline{D}^{w}_{Q,h}, \qquad \Delta E = E_3 - E_1,$$

where $\overline{D}_{S,h}$ is the mean sell-off gap. These quantities describe conditional differences between the selected samples. A causal interpretation would require additional identification assumptions.

Uncertainty is estimated using a circular block bootstrap within each comparison year. The resampling sequence contains every U.S. trading date in that year, including dates excluded from both comparison groups. Each date retains its previously calculated outcomes and eligibility indicators. Blocks of 20 consecutive U.S. trading dates are sampled with replacement, wrapping from the end of the year to its beginning when necessary, until the original annual sequence length is reproduced.

Each repetition recalculates the sell-off means, annual quiet-day means, annual event weights, and adjusted gaps. The change ΔE is calculated within the same repetition. The procedure uses 10,000 repetitions, with 95% percentile intervals calculated from the resulting distributions. The random seed is $42 + L$, where L is the block length.

4.5 Robustness Checks

Three sets of checks assess sensitivity to the analytical choices. First, the sell-off threshold is changed to −1.5% and −2.5%. Each sample uses the same calendar eligibility rule, and overlap clusters are reconstructed before estimating all three return-gap measures.

Second, each year represented in the main event sample is excluded in turn. Overlap clusters and bootstrap confidence intervals are recalculated for each reduced sample. The reported leave-one-year-out comparison focuses on the Day 1 gap.

Third, the quiet-day comparison is repeated using block lengths of 10 and 40 U.S. trading sessions. The underlying sample and point estimates remain fixed across block lengths, while the resampling distributions and confidence intervals vary.

The threshold samples are nested, and the year-exclusion samples share many observations. These checks assess sensitivity across related specifications. All reported confidence intervals are calculated individually, without adjustment for multiple testing.

Chapter 5 Results

5.1 Main Results

The calendar filter retains 65 of the 85 candidate U.S. sell-off events. These observations form 47 groups of overlapping event windows. Table 2 reports the mean KASE–HSI cumulative return gap at each horizon and its change from Day 1 to Day 3. All gaps and changes are expressed in percentage points.

The primary comparison is the change in the signed gap between the two horizons. Its estimate is −0.210 percentage points, with a 95% cluster-bootstrap confidence interval of [−0.995, 0.391]. The point estimate indicates a decline in the average signed gap, while the interval accommodates both negative and positive changes. Evidence about the direction of this change therefore remains inconclusive.

The Day 1 mean gap is 0.669 percentage points, with a 95% confidence interval of [0.345, 1.003]. Under the sample definition and inference procedure used here, this supports a positive average relative return for KASE compared with the HSI. By Day 3, the estimated gap is 0.459 percentage points, with a wider interval of [−0.441, 1.177], indicating considerable uncertainty about relative performance at that horizon. The cross-horizon conclusion follows from the interval for the change itself.

Table 2. Main KASE–HSI cumulative return-gap estimates

Measure	Estimate	95% cluster-bootstrap CI
Mean gap, Day 1	0.669	[0.345, 1.003]
Mean gap, Day 3	0.459	[–0.441, 1.177]

Change in gap, Day 1 to Day 3	–0.210	[–0.995, 0.391]

All values are percentage points. Estimates use 65 events in 47 overlap groups. Confidence intervals use 10,000 cluster-bootstrap repetitions. Throughout this chapter, intervals are reported individually without adjustment for multiple testing.

5.2 Comparison with Quiet Days

The quiet-day comparison uses 672 eligible observations from years represented in the sell-off sample. Annual quiet-day mean gaps are weighted to match the annual distribution of retained sell-off events.

On Day 1, the sell-off mean gap of 0.669 percentage points exceeds the weighted quiet-day gap of 0.063 by 0.606 percentage points. The 95% block-bootstrap confidence interval for this benchmark-adjusted gap is [0.209, 1.002]. This supports a positive Day 1 difference between the selected sell-off and quiet-day samples.

On Day 3, the estimated benchmark-adjusted gap is 0.324 percentage points, with a 95% interval of [−0.680, 1.303]. Its estimated change from Day 1 to Day 3 is −0.282 percentage points, with an interval of [−1.101, 0.485]. Both intervals contain zero, leaving the direction of the Day 3 adjusted gap and its cross-horizon change uncertain. The comparison is conditional on the sample selection and annual weighting rules. A causal interpretation would require additional identification assumptions.

Table 3. Sell-off gaps and the year-weighted quiet-day benchmark

Measure	Sell-off gap	Weighted quiet-day gap	Benchmark-adjusted gap	95% block-bootstrap CI for adjusted gap
Day 1	0.669	0.063	0.606	[0.209, 1.002]
Day 3	0.459	0.134	0.324	[–0.680, 1.303]
Change, Day 1 to Day 3	—	—	–0.282	[–1.101, 0.485]

All values are percentage points. Intervals use 10,000 circular block-bootstrap repetitions with blocks of 20 U.S. trading sessions. Adjusted gaps and changes are calculated from unrounded values, so displayed components may differ by 0.001 from their displayed difference.

5.3 Sensitivity to the Sell-Off Threshold

The Day 1 mean gap is positive under all three sell-off thresholds. Its estimate is 0.439 percentage points at the −1.5% threshold, 0.669 at −2.0%, and 0.815 at −2.5%. Each corresponding 95% cluster-bootstrap interval lies above zero.

More severe sell-off subsamples therefore have larger Day 1 point estimates. This pattern is descriptive because the samples are nested and differences between threshold-specific estimates have yet to be tested directly.

Across all three thresholds, the Day 3 point estimates are positive and the estimated changes from Day 1 to Day 3 are negative. All six corresponding intervals include zero. The uncertainty concerning these two measures persists across the tested event definitions.

Table 4. Return-gap estimates by sell-off threshold

S&P 500 daily return	Events	Clusters	Day 1 gap (95% CI)	Day 3 gap (95% CI)	Change, Day 1 to Day 3 (95% CI)
≤ –1.5%	117	78	0.439 [0.178, 0.689]	0.109 [–0.470, 0.632]	–0.330 [–0.820, 0.111]
≤ –2.0%	65	47	0.669 [0.345, 1.003]	0.459 [–0.441, 1.177]	–0.210 [–0.995, 0.391]
≤ –2.5%	40	26	0.815 [0.428, 1.298]	0.627 [–0.533, 1.473]	–0.187 [–1.218, 0.566]

Estimates and confidence limits are percentage points. Each sample uses the same calendar filter, with overlap clusters reconstructed for that sample. Intervals use 10,000 cluster-bootstrap repetitions.

5.4 Leave-One-Year-Out Results

Table 5 reports the Day 1 results after excluding each of the eight years represented in the main event sample. Estimates range from 0.497 to 0.776 percentage points, and all eight 95% cluster-bootstrap intervals lie above zero. The positive Day 1 finding thus persists after each individual year is removed.

Excluding 2020 produces the largest estimate, at 0.776 percentage points, with an interval of [0.326, 1.220]. Excluding 2022 produces the smallest estimate, at 0.497, and the interval with the lower endpoint closest to zero, [0.066, 0.883]. These differences show sensitivity in the magnitude and precision of the pooled estimate. Each row describes the remaining multi-year sample, so assessing variation across individual years would require a separate analysis.

Table 5. Leave-one-year-out Day 1 estimates

Excluded year	Events	Clusters	Day 1 gap	95% cluster-bootstrap CI
2018	55	41	0.645	[0.325, 0.977]
2019	61	43	0.733	[0.424, 1.077]
2020	45	37	0.776	[0.326, 1.220]
2021	60	42	0.586	[0.237, 0.923]
2022	47	33	0.497	[0.066, 0.883]
2023	64	46	0.683	[0.361, 1.021]
2024	61	43	0.649	[0.306, 1.007]
2025	62	44	0.765	[0.463, 1.089]

Estimates and confidence limits are percentage points. Overlap clusters are reconstructed after each year is removed. Intervals use 10,000 cluster-bootstrap repetitions.

5.5 Block-Length Sensitivity

Table 6 compares the quiet-day adjustment using circular block lengths of 10, 20, and 40 U.S. trading sessions. The point estimates are identical across these specifications because they use the same observations and weighting rule. Block length affects the bootstrap distributions and confidence intervals.

The Day 1 adjusted gap is 0.606 percentage points, and its confidence interval lies above zero under all three block lengths. The Day 3 adjusted gap is 0.324 percentage points, while its change from Day 1 is −0.282. The intervals for both measures include zero under every block length. The qualitative inference is therefore consistent across the three resampling specifications examined.

Table 6. Quiet-day benchmark sensitivity to block length

Block length	Day 1 adjusted gap (95% CI)	Day 3 adjusted gap (95% CI)	Change in adjusted gap (95% CI)
10 sessions	0.606 [0.216, 1.021]	0.324 [–0.678, 1.288]	–0.282 [–1.126, 0.489]
20 sessions	0.606 [0.209, 1.002]	0.324 [–0.680, 1.303]	–0.282 [–1.101, 0.485]
40 sessions	0.606 [0.233, 0.973]	0.324 [–0.566, 1.164]	–0.282 [–1.065, 0.440]

Estimates and confidence limits are percentage points. Block lengths are measured in U.S. trading sessions. Each specification uses 10,000 circular block-bootstrap repetitions within years.

Chapter 6 Discussion and Limitations

6.1 Interpretation of the Main Findings

The primary research question concerns how the signed KASE–HSI cumulative return gap changes from the first to the third local trading day following selected U.S. sell-offs. The estimated change is −0.210 percentage points, but its 95% confidence interval includes both negative and positive values. The direction of the average change therefore remains uncertain. A decline in the signed gap also has different implications depending on its initial position. For an event with a negative initial gap, a further decline increases the absolute separation between the markets.

The clearest finding is the positive average Day 1 gap. KASE's cumulative return exceeds the HSI's by an estimated 0.669 percentage points, with a confidence interval entirely above zero. The gap also exceeds the year-weighted quiet-day benchmark by 0.606 percentage points, with an interval above zero. Together, these estimates indicate stronger average relative performance for KASE on the first local trading day within the selected sample.

By Day 3, uncertainty is greater. The estimated gap remains positive, while its interval accommodates either market having the higher average cumulative return. Establishing the direction of the cross-horizon change requires inference on that change directly. The corresponding interval remains inconclusive in both the main and quiet-day-adjusted analyses.

6.2 Robustness and Possible Explanations

The positive Day 1 gap persists across the three sell-off thresholds and all eight leave-one-year-out samples, with every corresponding confidence interval above zero. The benchmark-adjusted Day 1 gap also has an interval above zero under each tested block length. These checks provide evidence of stability across the specific analytical choices examined.

The larger Day 1 estimates under more severe sell-off thresholds describe variation across nested samples. Assessing whether the differences between thresholds are statistically distinguishable would require a direct comparison. Similarly, leave-one-year-out estimates describe pooled samples remaining after each exclusion. Assessing individual years would require year-specific estimates and an explicit analysis of heterogeneity.

Several possible explanations deserve further investigation. Differences in industry composition may affect the indices' exposure to the news surrounding U.S. sell-offs. Domestic developments and common global news may also contribute to their relative performance. Trading activity and the timing of constituent price updates could influence how movements appear in daily index values. The current design estimates aggregate return differences, while identifying the contribution of each explanation would require additional data and controls.

6.3 Limitations

The main sample contains 65 events from 2017–2025, organized into 47 overlap groups. Cluster resampling preserves dependence within these groups, but dependence may also extend across groups through persistent market conditions. The block bootstrap similarly relies on the chosen blocks adequately representing relevant temporal dependence. Confidence intervals consequently depend on the suitability of these resampling procedures. All intervals are reported individually without adjustment for multiple testing.

Calendar alignment is implemented through daily dates. For each market, the filter counts U.S. trading dates from the baseline date, inclusive, to the Day 1 date, exclusive, and requires exactly one. The economic interpretation of this rule depends on the ordering of the exchanges' trading sessions. Historical opening and closing timestamps require separate verification. Furthermore, Day 3 follows each market's own calendar, so the elapsed time can differ between markets.

The measures use local-currency price indices. Dividends, exchange-rate movements, and transaction costs would affect investor outcomes measured on a total-return or common-currency basis. Index composition also differs across markets, meaning that country-level comparisons combine market characteristics with differences in the companies represented.

Price updating presents an additional limitation. Although the KASE series contains only one zero-return day, movements in an aggregate index can coexist with stale prices among some constituents. Evaluating this possibility would require constituent-level trading activity and price timestamps.

The observed return gap alone provides limited information about the speed of information incorporation.

Finally, both comparison samples depend on the event definition, calendar filter, and quiet-day exclusions. Annual weighting aligns their distribution across years, while within-year differences in economic conditions and news exposure may remain. The estimates therefore describe conditional associations. Causal attribution to U.S. sell-offs would require a design that addresses these competing influences.

6.4 Implications for Further Research

A priority is to verify historical exchange hours and use timestamped prices to construct comparable information windows. Intraday data could help distinguish movements occurring before the U.S. event close from those occurring afterward and show when subsequent changes emerge.

Constituent-level data would allow an examination of trading frequency, stale prices, and sector composition. Sector-adjusted comparisons could assess how much of the aggregate gap is associated with differences in index membership. Total-return indices and conversion into a specified base currency would extend the analysis to investor outcomes.

A longer sample could improve precision and support comparisons across market conditions. Future analysis could also directly test differences between sell-off thresholds and examine changes in absolute return gaps, with primary comparisons specified in advance and inference adapted to multiple testing where appropriate.

Chapter 7 Conclusion

This study compared the cumulative returns of the Hang Seng Index and Kazakhstan's KASE Index following sharp S&P 500 declines during 2017–2025. It examined whether the signed KASE–HSI return gap changed from the first to the third local trading day. The calendar-filtered sample contained 65 events, organized into 47 overlapping-window clusters for resampling.

The primary comparison remains inconclusive. The estimated gap declined from 0.669 percentage points on Day 1 to 0.459 on Day 3, giving an estimated change of –0.210 percentage points. The 95% confidence interval for this change, [–0.995, 0.391], accommodates both decreases and increases in the signed gap.

The clearest finding concerns Day 1. Its average gap of 0.669 percentage points has a 95% cluster-bootstrap confidence interval of [0.345, 1.003]. Positive estimates with intervals above zero persist across the tested sell-off thresholds and all leave-one-year-out samples. Relative to the year-weighted quiet-day benchmark, the Day 1 gap is higher by 0.606 percentage points, with a 95% block-bootstrap interval of [0.209, 1.002]. The benchmark-adjusted Day 3 gap and its cross-horizon change remain uncertain.

The study contributes a paired comparison of two markets following the same U.S. sell-off events, with explicit treatment of trading calendars, overlapping windows, and benchmark selection. Its findings describe conditional differences in local-currency price-index returns. Causal attribution

and investor outcomes incorporating dividends, currency conversion, and transaction costs require further analysis. Timestamped prices, constituent-level trading data, and total-return measures would help assess the mechanisms and scope of the observed Day 1 difference.